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REVIEW ARTICLE

Multi-modal, multi-step human–robot interaction method for natural interaction of service robots

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Abstract
Robots are becoming increasingly popular across various industries, including manufacturing, healthcare, agriculture, and personal services. This growing demand has spurred the development of robots with specialized abilities and applications, improving safety, productivity, and quality of life. This paper focuses on service robots with verbal and visual communication, which play a crucial role in human–robot interaction. We propose a novel method to enhance the accuracy of this communication by combining sound localization and image processing, enabling more natural interactions. Additionally, we aim to improve the service robot's mobility and interaction by adjusting its rotational speed to match human movement, thus enhancing adaptability and usability in diverse environments. This paper provides an overview of similar research efforts in human–robot interactions, details our proposed methodology, describes human–robot test environments, and presents experimental results, conclusions and implications for future research.

Keywords Voice recognition · Sound localization · Human–robot interaction · Image processing

1 Introduction

With continuous advancements in robotics, the use of intelligent service robots in our daily lives is on the rise. Furthermore, the recent global epidemic has had a profound impact on the development of remote communication technologies, resulting in rapid progress in this field. One of the significant advantages of having robots perform service roles is the ability to minimize the need for humans in the scene that previously required it. For instance, in shopping centers, robots can introduce themselves, collect used dishes in the cafeteria, deliver food, provide information, and even introduce products to customers [1, 2]. In the medical field, surgical robots can assist doctors in reducing the risk of complications during surgery and aid elderly and disabled patients with their daily needs. [3, 4] To interact with service robots, most of the existing studies and methods for human–robot interaction have used traditional device-based control methods. For example, industrial robots in factory environments may develop sophisticated control methods such as keyboards, mice, joysticks, and teach pendants [5]. However, these device-based interaction methods are not natural and greatly reduce the usability of the robots, especially service robots. Therefore, many studies have recently focused on natural interaction with robots [6, 7]. Joksimoski and Ong controlled the robot with body language such as hand gestures including sign language [8, 9]. However, they are often thought of as a secondary means of communication, as they limit the scope of expression [10]. Therefore, the most natural way of communication is verbal communication. So far, these studies have focused on how Kim, Sharifuddin, and Yang interacted with the robot by giving verbal instructions, in other words, voice recognition [11–13]. Additionally, these modules can function independently or in combination to enhance human–robot interaction. The key components include object recognition, verbal instruction processing, user detection, and gesture and gaze recognition [12, 14–19].

In this paper, we are focusing on service robots with verbal and visual communication, which plays a large role in service robots' interactions with people. To maximize the usability and effectiveness of communication, this study proposes a novel approach to improve the accuracy of its responses so that it interacts with people more naturally. The main contributions of this paper are,

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- We propose the multi-modal and multi-step HRI method to locate the person talking to a robot.
- We experiment with the average reaction time of a human to a robot and find out the natural reaction time.
- We control a robot to react to a human more naturally.

The paper is structured as follows: In Sect. 2, we thoughtfully reviewed earlier research studies, understanding their impact and contribution to our field. In Sect. 3, we describe our proposed main algorithm, which uses a combination of image processing and sound localization methods. In Sect. 4, we present the results of various experiments, along with a detailed explanation of how we created the test environment. Finally, in Sect. 5, we discuss the conclusions of this article and outline our plans for future work.

2 Related words

Substantial advancements have been made in human–robot interaction (HRI) to enable robots to perceive and interact naturally with humans. Earlier approaches focused on traditional device-based control methods, such as keyboards, joysticks, and teach pendants, which are widely used in industrial applications but lack the intuitiveness required for service robots [1, 5, 7].

Recent research has prioritized natural interaction modalities like voice, gestures, and facial expressions. Gesture-based control, including sign language recognition, has been explored to facilitate non-verbal communication. For instance, El-Komy et al. integrated computer vision and natural language processing to improve gesture and gaze recognition for multimedia robotics applications [7, 8, 20]. Similarly, Salim et al. proposed a comprehensive review of hand gesture techniques, highlighting their use in enhancing accessibility for people with hearing impairments [10, 21]. However, gesture-based approaches often require users to learn specific patterns, limiting their practicality for broader applications.

Voice recognition has emerged as a preferred method for natural HRI due to its intuitive nature. Studies such as Kim et al. and Sharifuddin et al. proposed voice-controlled systems to enhance mobility for disabled individuals, leveraging convolutional neural networks (CNNs) for improved speech recognition [11, 12]. Similarly, Tao et al. combined audio-visual features to enhance active speaker detection in multi-speaker environments [14, 16]. Despite these advances, challenges remain in distinguishing overlapping speech and determining speaker location.

Sound localization techniques have played a pivotal role in addressing these challenges. Valin et al. demonstrated the effectiveness of microphone arrays for sound source localization in dynamic environments, which laid the groundwork for robust robotic auditory systems [22, 23]. More recently, Pavlidi et al. employed real-time multiple speaker detection and Direction of Arrival (DOA) estimation using circular microphone arrays, significantly improving sound localization accuracy [24, 25]. Advances in machine learning have further optimized sound localization algorithms, enhancing their efficiency in real-world applications [22, 23, 25].

In parallel, computer vision techniques have seen rapid development in the context of HRI. The YOLO (You Only Look Once) family of object detection algorithms has become a standard for real-time facial and object recognition due to its high speed and accuracy [26–28]. As proposed by Liu et al., frame cropping and selective processing techniques have proven effective in reducing computational overhead while maintaining detection accuracy [6, 29].

Integrating multi-modal approaches for HRI has also gained attention in recent years. Griffin et al. introduced a framework combining sound localization with speaker identification and facial recognition for dynamic environments [24, 30]. Similarly, Khan et al. proposed combining voice and image processing to enable robots to respond effectively to diverse user commands [6, 14]. While these studies have made significant contributions, they often focus on isolated modalities or fail to address real-time constraints in multi-speaker environments.

Our work builds upon these advances by proposing a unified framework that combines sound localization, YOLO-based image processing, and mouth aspect ratio (MAR) analysis. This novel approach enables robots to accurately identify speakers, confirm speech activity, and reduce computational time through optimized frame processing. By integrating these techniques, we aim to enhance the naturalness and responsiveness of HRI systems, particularly in multi-speaker scenarios.

3 Multi-modal, multi-step human–robot interaction method for natural interaction

In today's era of rapid technological development, human–robot interaction has become an increasingly relevant topic. There are numerous methods available for handling human–robot interactions. Our objective is to develop a system that enables a robot to identify a person's location while they are speaking and to determine whether or not they are speaking. This will allow the robot to interact with people more naturally. To achieve this objective in an efficient and intelligent manner, we propose combining sound and image processing.

A robot's movement speed can be influenced by several factors, such as its purpose, design, and role. For instance, industrial robots used in manufacturing processes are designed to move quickly and precisely, allowing products to be assembled at high speeds. On the other hand,

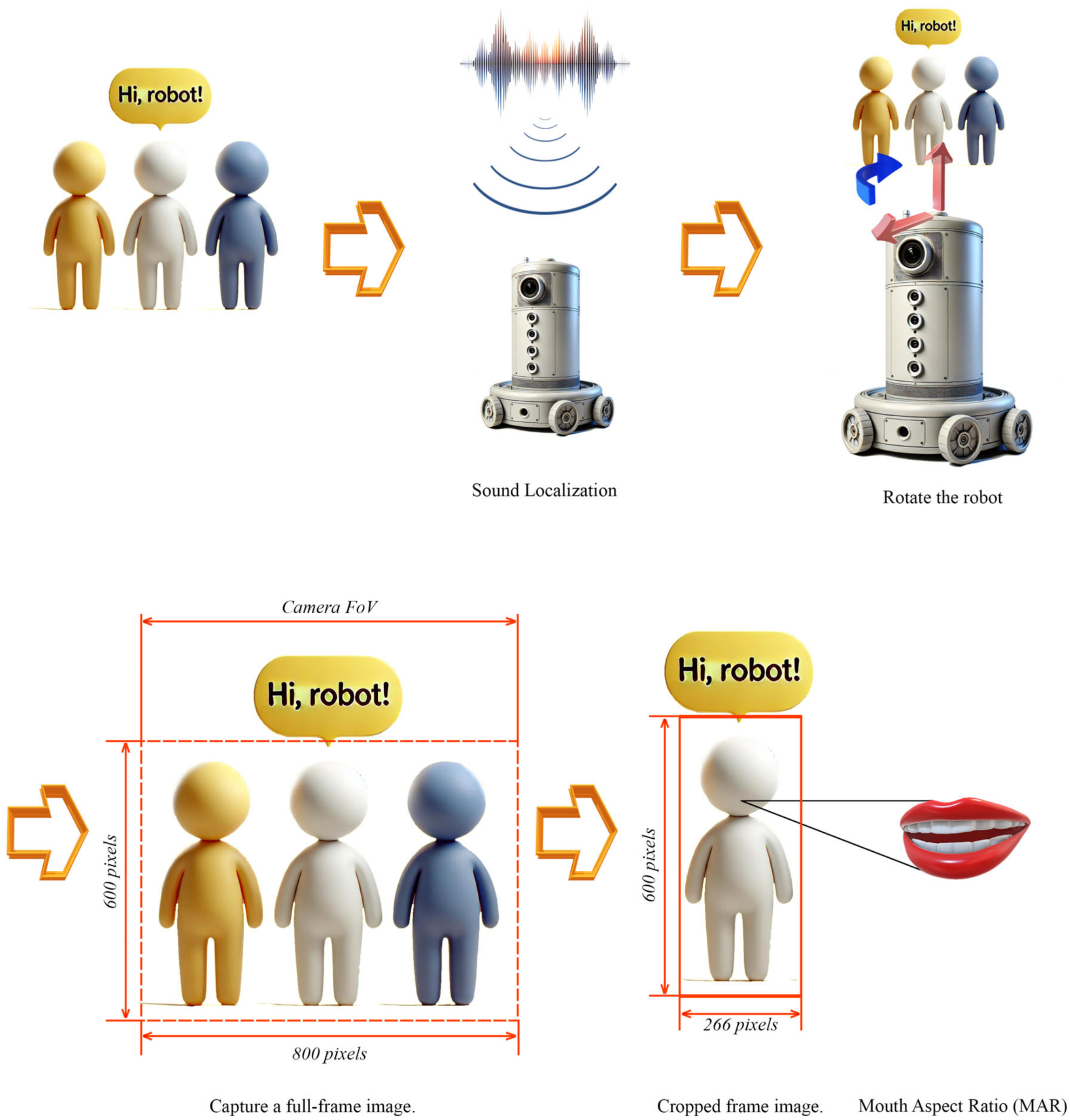


Fig. 1 The scenario of a multi-modal and multi-step HRI method

when it comes to service robots, response speed is not the most crucial factor in human–robot interactions. Therefore, the movement of service robots is generally slower than that of industrial robots. It is essential to consider the robot's proper speed and proper response time when developing service robots to ensure they can interact naturally with humans.

We took these factors into account when developing a service robot. We aimed to make the robot's movements as similar to human movements as possible. To archive this, we developed a novel multi-modal and multi-step HRI method to locate the person speaking to the robot using sound localization and image processing. After detecting the sound's direction, a face recognition algorithm was used to identify the person speaking by analyzing the mouth aspect ratio and the difference between frames. This method has the advantage of detecting the person speaking among a group of people and performing image processing only on that person.

By integrating these features, our service robot can move and respond to humans in a similar manner to humans while reducing image processing time. This can help improve the overall user experience and make the interaction more natural and intuitive.

Our proposed approach consists of three main components (Fig. 1). The first component involves capturing the sound and image data in the vicinity of the robot. The second component involves processing the captured data to extract relevant features such as the direction of sound and the presence of a human face. This component involves using machine learning algorithms to classify the features and determine the location and speech status of the person. The final component involves mouth aspect ratio (MAR) (Sect. 3.3).

3.1 Sound localization for speaker

A uniform circular array consists of m microphones (in Fig. 1, a robot has 6 microphone arrays), detecting S_g active sound source. Under the assumptions of the free field model, the signal captured by microphone m_i is given by:

$$x_i(t) = \sum_{g=1}^{j} a_{ig} s_g(t - t_i(\theta_g)) + n_i(t), \quad i = 1, \ldots, 6. \tag{1}$$

where:

- j number of speakers
- $x_i(t)$ is the signal received by the i^{th} microphone, m_i at time t.
- s_g represents the signal of the g^{th} sound source located at a distance q_s from the center of the microphone array.
- a_{ig} is the attenuation factor for the g^{th} source as received by m_i.
- $t_i(\theta_g)$ is the propagation delay for the signal from the g^{th} source to the i^{th} microphone.
- θ_g is the direction of arrival (DOA) of the g^{th} source with respect to the $x - axis$ (Fig. 2).
- $n_i(t)$ denotes the additive white Gaussian noise signal at the i^{th} microphone, which is uncorrelated with the sound source signals $s_g(t)$ and with all other noise signals from other microphones.

In the case of a single sound source, the relative delay in signals received between adjacent microphones designated as the microphone pair $\{m_i m_{i+1}\}$, is elucidated in references [20, 21].

Considering a uniform circular array of m microphones, we analyze signals from S_g active sound sources located in the array's far field. The signals are received at each microphone m_i as detailed in Eq.(1). Subsequently, we define the time delay between adjacent microphones m_i and m_{i+1} influenced by a sound source at direction θ_g, as

$$\tau_{m_i m_{i+1}}(\theta_g) \triangleq t_i(\theta_g) - t_{i+1}(\theta_g) = \frac{l sin(A + \frac{\pi}{2} - \theta_g + (i-1)\alpha)}{c} \tag{2}$$

as illustrated in Eq.(2), where:

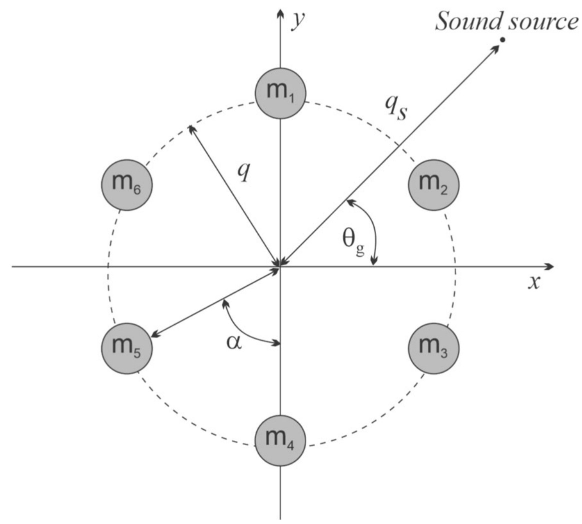


Fig. 2 Circular microphone array configuration. The microphones are m_i, $i = 1, 2, 3, \ldots 6$ and sound sources are S_g

- α is the angular separation between adjacent microphones, given by $\frac{2\pi}{m_i}$,
- l is the distance between adjacent microphones, calculated as $2qsin(\frac{\alpha}{2})$,
- A represents the obtuse angle formed by the chord connecting microphones m_1 and m_2 with respect to the $x - axis$, computed as $\frac{\pi}{2} + \frac{\alpha}{2}$,
- c stands for the speed of sound,
- q denotes the radius of the microphone array.

These relations are established in Eq.(3). Importantly, while the DOA θ_g in Eq.(1) is defined concerning the $x - axis$, reference [20] defines it relative to a line perpendicular to the chord formed by the m_1, m_2 microphone pair. Moreover, all angles in Eq.(2) and (3) are measured in radians. To estimate the number of active sound sources, S_g, and the respective DOAs θ_g, we process the mixture of source signals x_i, taking the known array geometry into account. Furthering our analysis, A. Griffin and A. Karbas define the phase rotation factors according to [20, 23]:

$$G^{\omega}_{m_i \to m_1}(\theta) \triangleq e^{j\omega\tau_{m_i \to m_1}(\phi)} \tag{3}$$

where the relative delay difference, $\tau_{m_i \to m_1}$, is given by $\tau_{m_1 m_1}(\phi) - \tau_{m_i m_{i+1}}(\phi)$ with $\tau_{m_i m_{i+1}}$ evaluated based on Eq.(2). Here, ϕ is in the range $[0, 2\pi]$ (in radians) and ω belongs to the set Ω. Building on this, A. Karbas and D. Pavlidi calculate the Circular Integrated Cross-Spectrum (CICS) using these phase rotation factors, as defined in [20, 23]:

$$CICS^{(\omega)}(\phi) \triangleq \sum_{i=1}^{M} G^{(\phi)}_{m_i \to m_1}(\phi) G_{m_i m_{i+1}(\phi)} \tag{4}$$

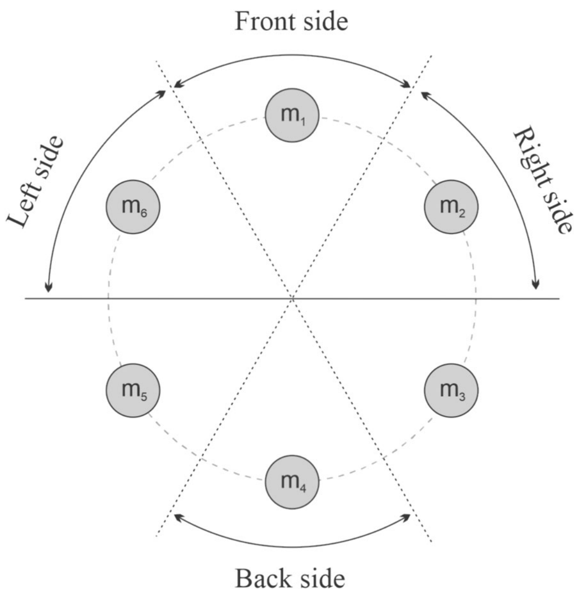


Fig. 3 The circular microphone array used in the experiment

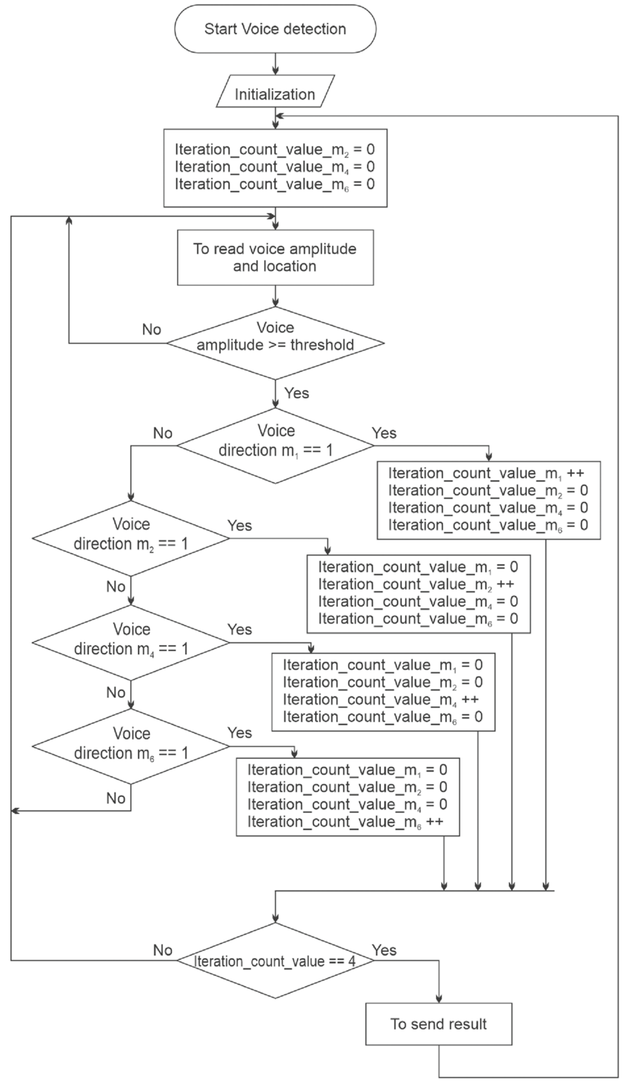
To estimate the DOA associated with the frequency component ω in a single-source zone within the frequency range Ω, D. Pavlidi use the method proposed in [23]:

$$\hat{\theta}_\omega = \arg \max_{0 \le \phi < 2\pi} \left| \mathrm{CICS}^{(\omega)}(\phi) \right| \tag{5}$$

We hone our focus on "strong" frequency components within each single-source zone to enhance the accuracy of DOA estimations. Following the approach of D. Pavlidi and A. Griffin, we exclusively utilize the frequency ω_i^{max}, corresponding to the peak component in the cross-power spectrum of the microphone pair $\{m_i, m_{i+1}\}$ within a single-source zone, thereby deriving a solitary DOA for each zone as depicted in references [22, 23].

The value that exceeded the predefined threshold we set, or the position of the microphone, was divided into four directions. The microphones and their corresponding numbers are shown in Fig. 3. The front of the robot is represented by m_1. m_2 and m_6 correspond to the left and right sides of the robot at 90 degrees, respectively, while m_4 represents the back side or 180 degrees.

In this paper, we employed a circular microphone array to ascertain the direction of sound. This was achieved by analyzing the peak frequency value of the signal captured by each microphone within the array. We established a threshold based on the magnitude of the signal from the microphone. If this threshold is surpassed, it is inferred that an individual within the detection area is speaking. A comprehensive description of the operational algorithm is provided in algorithm 1. The system we developed utilizes a total of six microphones to capture the signal and subsequently relay sound localization data to a robot. Upon receipt of this position data, image processing is initiated, directing the robot toward the detected sound source.



Algorithm 1: Flowchart of the algorithm sound localization.

3.2 Image processing for human detection

After detecting the sound and direction, like a precious subsection, a robot needs to recognize humans. Thus, we use an image processing method. When image processing was performed, images were received by "Pi camera V2." Camera specification is 8 Megapixels, 3280 × 2464 still picture resolution, Support 1080 p30, 720 p60, and 640x480 p90 video record. In this paper, we have employed the You Only Look Once (YOLO) [24, 25] architecture for the object detection component within our image processing pipeline. The network architecture of this model comprises 24 convolution

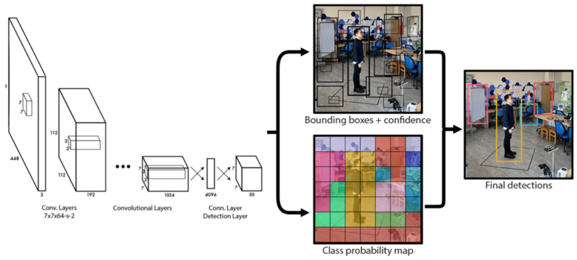


Fig. 4 The detection pipeline of YOLO: the input image is divided into a S x S grid where the bounding boxes are simultaneously predicted with corresponding confidence and class probability values [24]

Fig. 5 Human face recognition with full frame and cropped frame

layers, followed by two fully connected layers. These convolution layers are responsible for feature extraction, while the fully connected layers predict the bounding box locations and their associated probabilities. The system begins by segmenting the input image into an S x S grid. These grid cells are associated with two bounding boxes and their respective class confidences. As a result, up to two objects can be detected within a single cell. If an object spans across multiple cells, the prediction is anchored to the central cell of that object. During the training phase, a bounding box without any object receives a confidence value of zero. On the other hand, a bounding box that encircles an object is assigned a confidence value equivalent to the intersection-over-union (IoU) score between the predicted bounding box and the ground truth box (Fig. 4).

Table 1 Comparative overview of image processing methods (seconds)

Methods	Image 1	Image 2	Image 3
Existing method	0.15	0.15	0.17
Our method	0.06	0.07	0.07

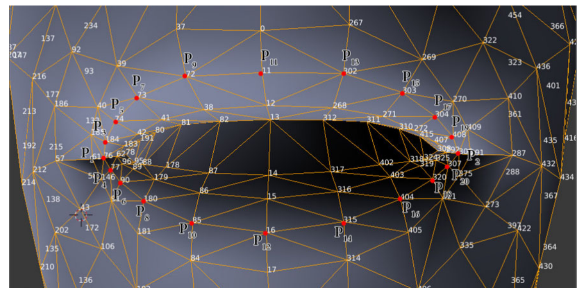


Fig. 6 Facial landmark points with $p_1 \sim p_{20}$ of the mouth [27]

One of the challenges in image processing is computational speed. A solution to this problem is frame splitting, which is especially effective when the location of sound is known, enabling the robot to focus in that direction. This simplifies image processing as the object will be centered in the camera's field of view. When processing images, a 3:1 vertical crop is recommended instead of full-frame processing.

For instance, processing a frame with a ratio of 800x600 (Fig. 5a, c, e) would be reduced to a frame with a ratio of 266x600 (Fig. 5b, d, f), thereby halving the image processing time. Table 1 provides a comparative overview.

3.3 Mouth aspect ratio (MAR)

We employed the OpenCV library [26] to load video captures and convert them into three-dimensional matrices. Using the Facial Landmark model, we were able to extract essential points on the human face, including specific points pertaining to the lips. With precise facial recognition, we isolated the mouth aspect ratio (MAR) [27, 28, 31] area and computed the Euclidean distance between these key points. To elaborate, we calculated the sum of the three-dimensional Euclidean distances [29] for the vertical points and then divided this sum by twice the three-dimensional Euclidean distance between the two horizontal points. The distances for all these points were determined using the Euclidean distance.

In Fig. 6, $p_1 \sim p_{20}$ indicates the twenty facial landmark points of the mouth. MAR, mouth aspect ratio, is defined below:

$$MAR = \frac{\sum_{i=3}^{19} Distance\,(p_i, p_{i+1})}{2 * Distance\,(p_1, p_2)}, \tag{6}$$

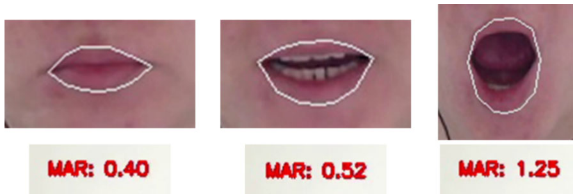


Fig. 7 MAR's for Various mouth opening

which is the sum of all the three-dimensional Euclidean distances of vertical points' will be divided by twice the three-dimensional Euclidean distance of the two horizontal points. The MAR value is at the peak for the fully open mouth and the value becomes zero for the fully closed mouth. So, increasing MAR values indicate that the mouth is gradually opening and the value is almost zero for a completely closed mouth. The twenty points ($p_1 \sim p_{20}$) for the MAR from 468 MediaPipe Face Mesh landmarks are 76, 306, 77, 184, 90, 74, 180, 73, 85, 72, 16, 11, 315, 302, 404, 303, 320, 304, 307, 408, respectively [30, 32].

In Fig. 7, the illustration depicts the dynamic movement of the lips and the changing ratio as a person speaks. We collected these distances and computed the average by comparing 16 consecutive data points in a 2x8 ratio. If the average value is below the threshold (Th), it indicates that the person is not speaking. Conversely, if the average value exceeds the threshold, it indicates that the person is speaking.

In detecting moving objects in videos captured by stationary cameras, the frame difference method is often considered the simplest. This is due to its fast detection speed, ease of hardware implementation, and widespread use [33, 34].

Using the frame difference method, unchanged parts of the scene are eliminated in the difference image, while changed parts remain. These changes can be a result of movement or noise. Consequently, a binary process is applied to the different images to distinguish between moving objects and noise. Additionally, connected component labeling is used to identify the smallest rectangle encompassing the moving objects.

For the purpose of calculating the threshold in the binary process, noise is typically assumed to be Gaussian white noise. Based on statistical theory, it is rare for any pixel to exhibit a dispersion more than three times its standard deviation. As such, the threshold is calculated accordingly.

$$Th = u \pm 3\sigma \tag{7}$$

While u is the mean of the difference image, σ is the standard deviation of the difference image.

This component extracts the lip region from a human face in an image frame provided by the image processing unit. Subsequently, the extracted frame is resized to dimensions of 400x400 pixels to prepare it for comparison. The same procedure is then applied to the next frame. Following this, the first and the subsequent frames are compared. If the two frames are identical, it is concluded that the person is not speaking. Conversely, if there are differences between the frames, it is assumed that the person is speaking. The mouth aspect ratio of the differences between the frames serves as an indicator of whether the person is vocalizing.

3.4 Voice detection

After the robot comes to a stop based on the position received from the sound localization algorithm 1, human detection (Fig. 5) and mouth aspect ratio detection (Fig. 7) the voice detection system is activated. During this process, the voice label is set to 1 if a continuous signal is received from microphone m_1 (see Fig. 3). However, if no signal is detected from microphone m_1, the voice label is set to 0, indicating that no one is speaking in that direction.

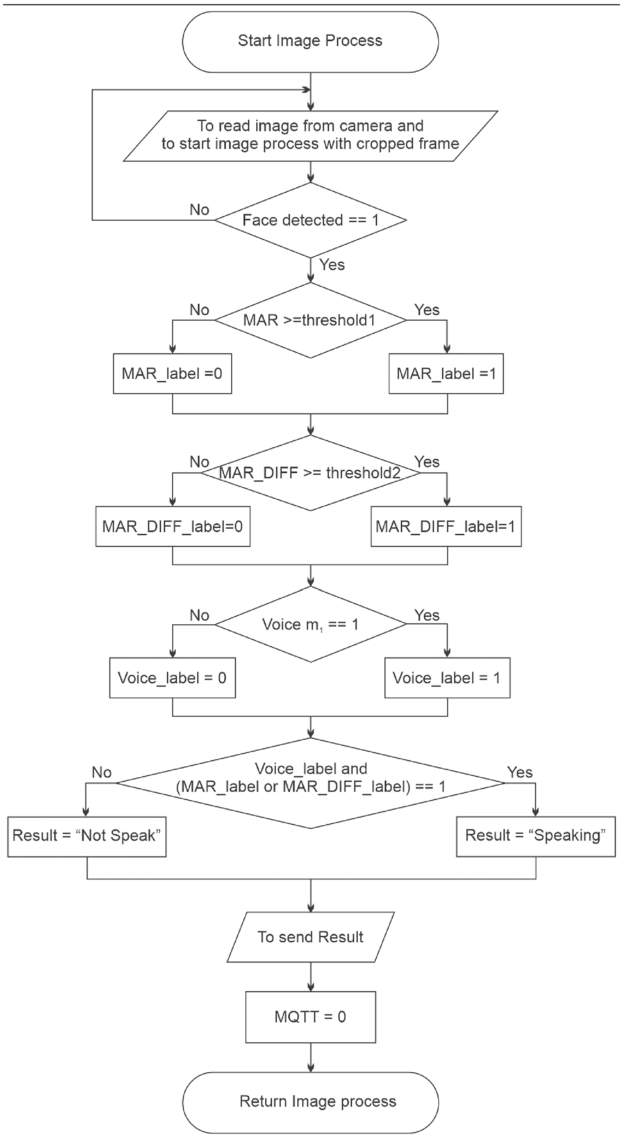
Thus, our proposed algorithm 2 can be used to determine whether a person is speaking or not. When a person is only moving their lips, MAR-label will be 1 and MAR-DIFF-label will be 1, but Voice-label will be 0 indicating that the person is not speaking. Furthermore, if the person is speaking, MAR-label will be 1, but MAR-DIFF-label will be 0 and Voice-label will be 0 as well, so the result will not be speaking. However, if the person is speaking, the MAR-label will be 1, MAR-DIFF-label will be 1, and Voice-label will be 1, then the result will be speaking.

4 Experiments

This section provides an overview of two types of experiments designed to synchronize robot movements with those of humans. The primary aim of these experiments is to align the robot's speed and reaction time with human capabilities. In the first experiment, we conducted a thorough analysis of various aspects of human movement, such as acceleration, deceleration, and reaction time. This analysis led to the development of a real-time model that enables the robot to replicate these characteristics accurately. The objective of this synchronization is to improve the robot's performance in dynamic environments, particularly in tasks that require effective interaction with humans.

4.1 Experiment on human reaction time

In the first experiment, our aim was to measure the time spent by humans when they turn toured a sound source. Approximately 200 individuals between the ages of 15 and 35 participated in an experiment in which the direction of the sound was divided into four parts, and time was measured while the participants turned in different directions.



Algorithm 2: Flowchart of the algorithm for human speech identification based on mouth aspect ratio, frame-to-frame difference, and audio signal.

For our experiments, we developed the sound emulation system Fig. 9 shows the hardware implementation like Fig. 8 to measure the duration it takes for a person to shift their gaze toward a direction upon hearing a sound. The experiment took place in a room measuring 6x8 ms (Fig. 10). Speakers were positioned in four directions, around the test object each, at a distance of 2.5 ms. Figure 11 illustrates how the object carries out its actions, captured by a camera mounted on the ceiling of the testing environment. Figure 12 shows the average turning time of all tested persons.

In the first experiment, our aim was to measure the time spent by humans when they turn their bodies toward a sound

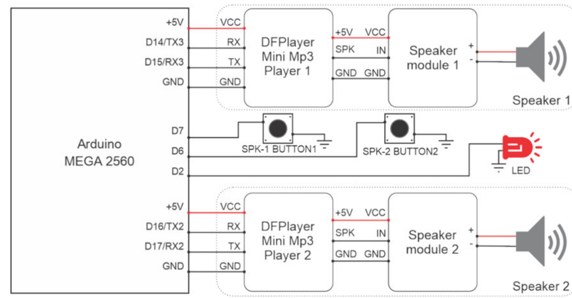


Fig. 8 The block diagram of the sound emulation system

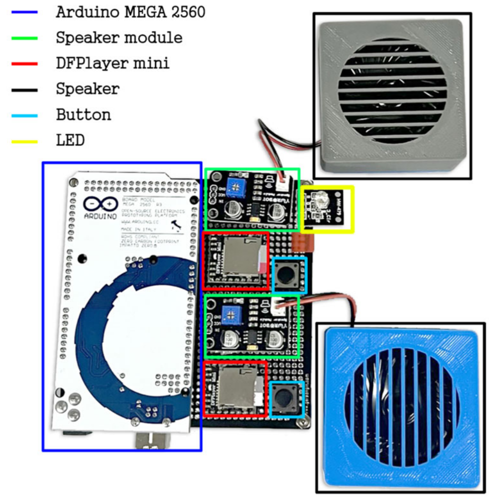


Fig. 9 Sound emulation control board

source. For this experiment, we conducted the test under the condition that the person turns their whole body in the direction of the sound.

To create the test environment, four 8 Ohm speakers were used along with an "Amplifier module." The sound was played in "mp3" format using the "DFPlayer mini" sound module. The control system was created using two "Arduino MEGA 2560" boards, with each board controlling two "DFPlayer mini" modules to play "mp3" files. Figure 8 depicts the block diagram of the connection, while Fig. 9 shows the control panel assembled according to the block diagram.

In addition, to prepare the experiment environment, four speakers are placed on the four corners. When the person stands under the camera placed on the ceiling of the room, the start button on the control board is pressed by the operator. Then, one of the speakers will be randomly sounded. After that, turning time to the sound direction of a person is captured on video.

Afterward, the video file is edited using "Premiere Pro (Adobe Inc., USA)" software, and the lookback time of each person is converted to seconds using a "timecode calculator."

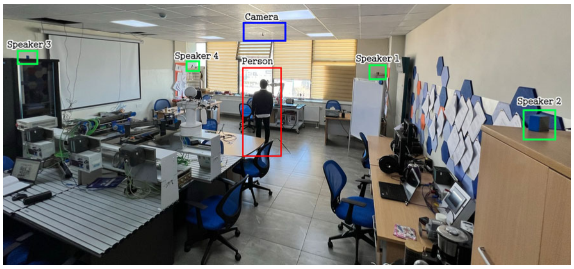


Fig. 10 Experimental environment

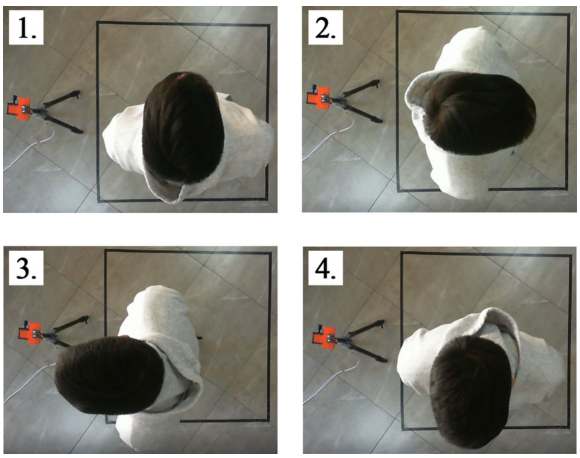


Fig. 11 The first scene is the front. In the second and third scenes, the person is turned 90 degrees to the right or left. In the fourth scene, the person is turned 180 degrees or facing the back side

Figure 10 depicts the testing room, equipment, and the tested subject (a human).

During data collection, videos were recorded with a camera positioned above the participants. Figure 11 displays a snapshot of the video data of the test subjects.

From the graph (Fig. 12), it can be seen that the average time of clockwise 90-degree movement of the person is 2.57 s and the average of 90 degrees counter-clockwise movement is 2.54 s, so the difference between them is 0.03 s. In addition, the average turning time of a 180-degree clockwise movement is 3.25 s, and the average time of a 180-degree counter-clockwise movement is 3.22 s, the difference between them is 0.03 s.

The purpose of the first experiment is to match the turning speed of a robot with humans by finding the optimal movement time of a human.

4.2 Setup and experiment for robot

In this experiment, we utilize the turtlebot3 depicted in Fig. 13 and integrated the previously discussed experimental environment in Fig. 14. Our robot is controlled by two Raspberry Pi4 units. The first Raspberry Pi4 controls the robot's main body and processes the position signal emanating from the Mic Array [35]. The second Raspberry Pi4 is responsible for image processing, utilizing images taken by the "Pi camera V2."

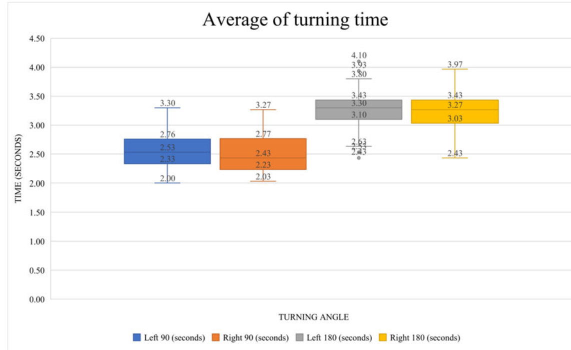


Fig. 12 A graph of the average turning time of all persons tested

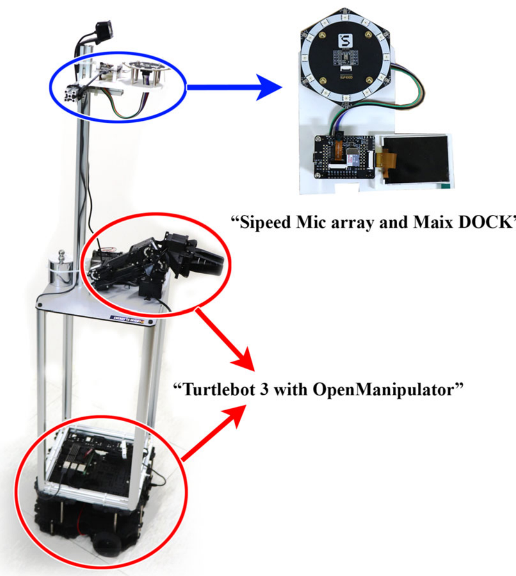


Fig. 13 "TurtelBot 3" testing robot with microphone array and Maix DOCK [36]

Communication between the two Raspberry Pi units is facilitated through the "MQTT protocol." The primary function of the main Raspberry Pi is to determine the sound's direction based on the input from the Mic Array. Once identified, it transmits a signal to the second Raspberry Pi. Upon receiving this signal, the second Raspberry Pi commands the turtlebot to turn and face the sound's source. The optimal rotation speed of the robot wheel was set to 210 RPM by carefully analyzing the time taken by the human to rotate from the experiment results in section 4.1. This adjustment significantly improved performance as the robot completed a 90-degree turn 0.47 s faster than a human. However, it is worth noting that this adjustment increased the time required to turn 180 degrees, causing a delay of 0.07 s. The results of these tests are shown in Table 2.

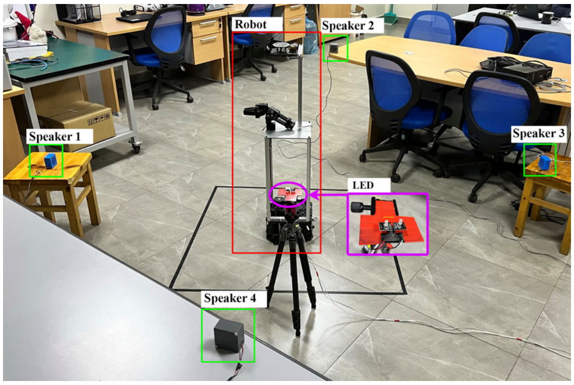


Fig. 14 Robot experimental environment

Fig. 15 Test environment and robot position

Table 2 Comparison of human and robot rotation times (second)

Experimental type	90 degrees	180 degrees
Human rotation time	2.53	3.33
Robot rotation time	2.06	3.26

The instruction to initiate image processing is then dispatched to the second Raspberry Pi via the "MQTT broker." Notably, rather than processing the full frame, the system only processes a cropped frame this commences once the direction of the speaking individual has been pinpointed (as illustrated in Fig. 5).

The image processing of our proposed method takes 0.15 s to process one loop with a full-frame (Fig. 5a, c, e). To reduce this time, based on the results of previous experiments, the robot can determine where the person is speaking with the help of the Mic array and look in the direction of the person speaking. When looking at a person, the camera's angle of view is focused on the person, so by dividing the incoming image into 3 equal parts along the vertical axis, cutting out one of them, and performing image processing on that frame (Fig. 5b, d, f). The time of image processing with the cropped frame was 0.07 s.

An experiment was conducted to test the ability of a robot to detect which person would speak when three individuals entered a test environment and interacted with it by looking at the robot. Figure 15 illustrates the position of the humans and the robot position in the test environment.

Conducted a series of tests utilizing the recommended algorithms and a split-screen method for enhanced analysis. This experiment assesses and determines whether an individual is actively speaking accurately.

In Fig. 16, frames 1–3 depict the initial result displayed as "True" on the screen, indicating that the person is speaking, as determined by Algorithm 1 and Algorithm 2. However, in the 4th frame, the person begins speaking, resulting in the displayed outcome changing to "False." Subsequently, frame 5 shows the result as "True" because the person continues to speak. In frames 6-7, the result shifts to "False" as the person stops talking.

Upon detection that another person begins speaking from a different location, the algorithms send the position, prompting the TurtleBot to focus on the new speaker and initiate image processing. Frames 8–11 show the displayed result as "True." Conversely, in frames 12–14, the result is "False" as people resume talking.

Frames 15–17 illustrate the algorithm detecting and transmitting the position, resulting in a "True" outcome. This indicates that the TurtleBot is directed to focus on the object and commence image processing. Finally, frames 18–20 present the algorithm's output as "False."

We conducted it three times, and the results are shown in Table 3.

The sound localization test only identifies the direction of the sound source, prompting the robot to point accordingly. As a result, determining the number of people in the environment is impossible, leading to inconclusive outcomes in tests involving two or three people. While the acceptable range is less than $\pm 15°$, the sound localization test demonstrates an accuracy of $\pm 5°$, which is significant given that the location definition is confined to the sound direction.

However, experiments combining sound localization with YOLO demonstrate its effectiveness in identifying the origin of sounds and counting the number of people through object detection. This approach significantly reduces the error to $\pm 1° \sim \pm 1.5°$, well within the acceptable range.

In our proposed method, the position is determined through sound localization, the image through YOLO, and the identification of a speaking individual by analyzing the mouth aspect ratio. This approach minimizes angle error. Furthermore, dividing the frame into thirds halves the image processing time, providing an advantage over the previous two methods.

Fig. 16 The detection of the speaking person is displayed on the screen

5 Conclusions and future work

In this paper, the purpose is to find the condition in which the service robot is closest to the time it takes for a human to turn around. And determining the position of human to reduce the time required for image processing and identify the person speaking. To determine the time required for a person to turn 90 degrees, more than 200 humans were tested, with an average time of approximately 2.54 s. Our proposed methodology seeks to enhance the robot's responsiveness to human interactions. Initial testing revealed the robot's turning time to be 2.07 s, aligning closely with human movement patterns. Subsequently, we optimized efficiency by halving the image processing time, resulting in a refined total reaction time of 2.1 s for the robot. This reduction brings the robot's response time more in line with human reaction time, signifying a substantial improvement in its adaptability to human-centric scenarios.

In our ongoing efforts to enhance our service robot design, we focus on developing advanced machine learning models for image processing and leveraging OpenAI for sound processing. Additionally, we will plan to implement citizenship and language recognition systems for our service robots. These advancements will improve the control mechanisms of the robots and enable them to perform tasks more effectively with human guidance.

Table 3 Experimental results (time (seconds)/degree)

Experiments (I−III)	SL[1] only	SL[1]+YOLO[2]	Our method
One person	0.01/±5°	0.15/±1°	0.03/±0.5°
Two person	-	0.11/±1°	0.04/±0.5°
Three person	-	0.15/±1°	0.03/±0.5°
One person	0.03/±5°	0.16/±1.1°	0.05/±0.3°
Two person	-	0.11/±1.5°	0.05/±0.4°
Three person	-	0.15/±1.2°	0.05/±0.2°
One person	0.02/±5°	0.17/±1.3°	0.04/±0.5°
Two person	-	0.11/±1°	0.03/±0.5°
Three person	-	0.13/±1.1°	0.04/±0.4°

[1] Sound localization
[2] You Only Look Once

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